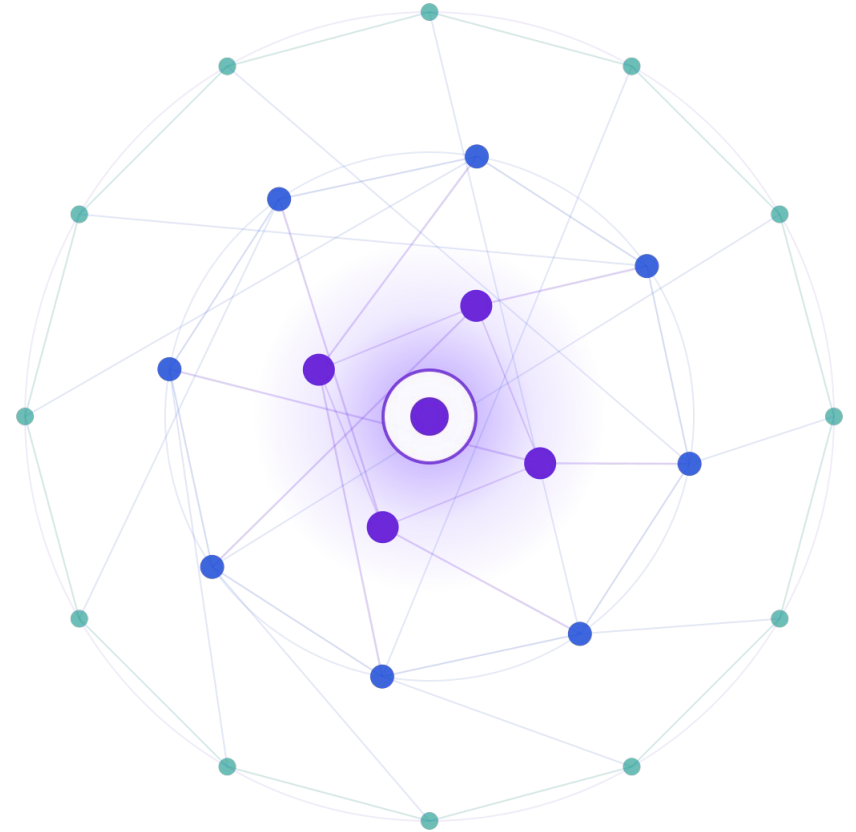


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 4
- Team Name - Binary Bits
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- Submission Github link - <https://github.com/Aakarsh-Kumar/prism-live-rag>



Theme & Problem Statement

Grounding Answers While the Conversation Is Still Unfolding

Spoken requests arrive incrementally, with pauses, revisions and late constraints. A grounded assistant must decide when enough intent is available to retrieve evidence and when it should wait.

WHY THIS MATTERS:

- Waiting for the complete request delays retrieval.
- Searching every partial wastes compute and retrieves premature evidence.
- Revisions can invalidate previously relevant passages.

CORE CHALLENGE:

- Retrieve early without committing to an unstable interpretation.

WHERE EXISTING RAG FALLS SHORT

The problem isn't just finding information. It's knowing when the user's request is ready to search.

Solution

Conventional two-step RAG:
Retrieves context, then produces an answer; Needs additional control for evolving ASR hypotheses.

What it does

Agentic RAG: Uses an LLM to decide when and how to retrieve
Variable latency; does not inherently enforce chunk-level timing.

Limitation for task

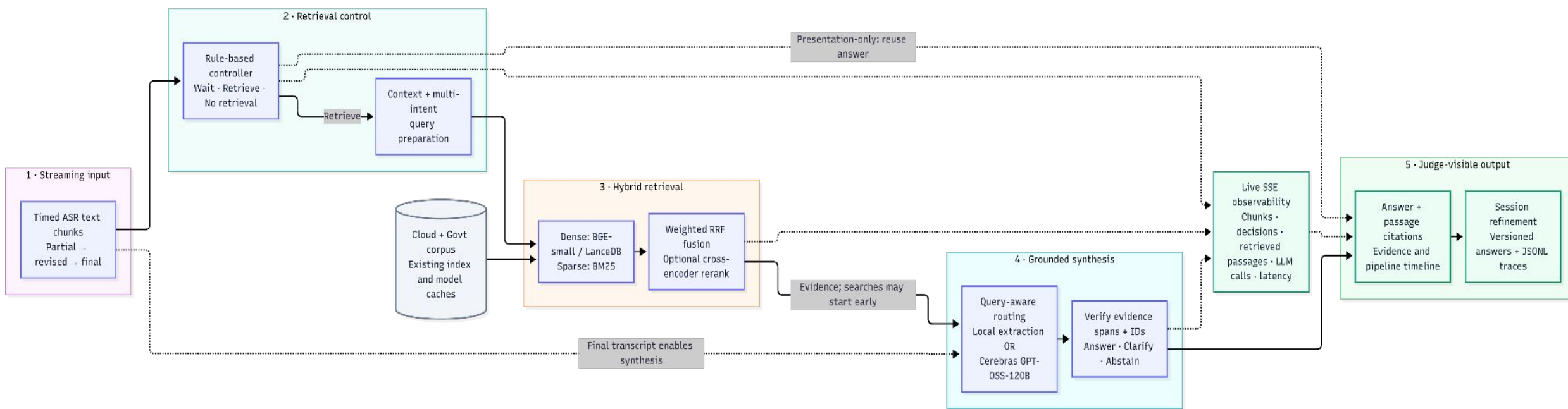
Dense-only retrieval: Finds semantically similar passages
Can miss exact terminology; achieved 68% Success@5 in our baseline.

Common Gap: Retrieval and answer generation alone do not manage partial speech, superseded hypotheses and late constraints.

Approach reference: LangChain retrieval architectures. Dense-only result: internal 50-query benchmark.

Our Solutions & Architecture Diagram

An **observable streaming RAG engine** that starts retrieval on stable partial transcripts and returns evidence-backed answers after the final hypothesis.



KEY COMPONENTS:

Controller: manages retrieval timing. ; **Decomposer:** separates independent intents. ; **Hybrid retriever:** fuses and reranks evidence. ; **Grounding layer:** validates source spans and passage IDs.

Demo & Product Walkthrough

A resident asks: “Who do I call about missed garbage collection, and what should I do if someone takes my recycling cart?”

Product Walkthrough:

Step 1 — Select and replay : Choose the compound query from the 200 dashboard scenarios.

Step 2 — Watch the request unfold : Transcript chunks arrive at intervals; controller decisions appear alongside them.

Step 3 — Inspect the evidence path : See subqueries, retrieved passages, ranking signals and actual provider-call status.

Step 4 — Inspect the result : Open passage citations and review the answer—or an explicit uncertainty outcome.

Demo: [Demo Link](#)

Tools and tech stack used

FRONTEND:

HTML, CSS, JavaScript : dashboard and evidence inspection

Server-Sent Events: live pipeline updates

BACKEND:

Python 3.11–3.13: streaming pipeline and CLI

Stdlib HTTP server: dashboard APIs

AI / LLM:

Cerebras GPT-OSS-120B: conditional decomposition and evidence selection

BGE-small: dense embeddings
Cross-encoder: optional candidate reranking

RAG / RETRIEVAL:

BM25 + weighted RRF: sparse retrieval and fusion

DATABASE / STORAGE:

LanceDB: dense index

JSONL: scenarios, checkpoints and execution traces

INFRASTRUCTURE / DEPLOYMENT:

Docker + Compose: CPU deployment

ONNX Runtime: CPU / optional local GPU inference

OTHER TOOLS:

pytest: regression testing

Git / GitHub: source and version history

Impact & Use case

PRIMARY USERS:

Support agents, service-desk operators and conversational-assistant developers.

USE CASE 1 - Technical Support

Retrieve documentation while a customer describes a cloud-service problem.

USE CASE 2 - Public-Service Assistance

Separate related municipal questions and expose the supporting source passages.

USE CASE 3 - Conversational Refinement

Apply a late constraint without unnecessarily replacing unrelated answer content.

MEASURABLE IMPACT:

90.77% early search dispatch: 118/130 eligible CPU replay cases.

+12 percentage points Success@5 versus dense-only retrieval with reranking.

200/200 validated execution traces in the saved CPU coverage run.

BROADER IMPACT:

Potential to overlap retrieval with speaking time and make evidence decisions inspectable.

End-to-end user latency savings have not yet been established.

Innovation highlights, results and limitations

INNOVATION HIGHLIGHTS:

Temporal retrieval: decisions operate on evolving transcript hypotheses.

Selective LLM assistance: eligible queries use Cerebras; others stay local.

Verifiable extraction: selected spans and passage IDs are checked against evidence.

RESULTS:

Success@5: 80% - hybrid + rerank.

Mean Recall@5: 58.49% - same 50-query GPU experiment.

Trace coverage: 200/200, with zero invalid citation IDs - saved CPU audit.

CURRENT LIMITATIONS:

Simulated ASR input; answers are text streams, not token-streamed.

CPU offline audit: 59/100 answerable cases were uncertain or uncited.

QUALITATIVE RESULT:

Session checks demonstrated presentation-only reuse without retrieval and delta-only refinement on a real corpus case.

What's next

IMMEDIATE:

Finish private asset delivery and verify a fresh-machine Docker launch.

Re-audit answerable abstentions after recent grounding fixes.

SHORT TERM:

Add a microphone / ASR adapter preserving revisions and timestamps.

Replace lexical multi-intent scoring with independent semantic review.

LONG TERM:

Introduce validated provisional answers and cancellable generation.

Measure concurrent-session throughput and speech-to-answer latency.

END GOAL:

A conversational assistant that prepares evidence while listening and updates answers safely as intent changes.

Brownie points slide (differentiation)

1. Inspect the execution

Visible chunks, decisions, retrieved passages, citations, and provider calls, not just a chat response.

2. Component choices

Have measured justification: Matched retrieval ablations compare dense, hybrid, and reranking configurations.

3. Auditable corpus

122,049 raw passages are preserved; retrieved citations retain original passage IDs.

4. Follow-ups preserve state

Presentation requests reuse answers; refinements retain version lineage and target delta evidence.

TECHNICAL MOAT:

Coordinating partial transcripts, evidence freshness, and versioned refinement without hiding unsupported claims.

ONE-LINE TAKEAWAY:

Not just an answer box, a visible, evidence-driven conversation pipeline.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N) - Y
- README with reproducible setup instructions (Y/N) - Y
- Demo video, max 5 minutes (YouTube or Drive link)- Y
- Presentation file (PPT or PDF) (Y/N) - Y

Thank you